Keiji AI

BioDSA Analysis Report

Report Generated:	2026-09-01 20:15:55
Total Iterations:	8
Code Executions:	0
Artifacts Generated:	0
Total Input Token Usage:	103572
Total Output Token Usage:	5288

User Query

Among Cas9-disrupted loci in human neural stem cells, what fraction of disruption phenotypes were apparent after 4 cell divisions? select from 61% of enhancer loci,31% of gene loci,31% of enhancer loci,61% conserved region loci,31% conserved region loci

Agent Exploration Process

The following shows the agent's reasoning and exploration process. Code blocks are truncated for readability - full implementations are in the Supplementary Materials section.

Step 2: AI

Tool call: go_breadth_first_search

Tool call input: {'search_rounds_budget': 4, 'knowledge_bases': ['pubmed_papers'], 'search_target': 'Find the PubMed paper and exact reported result about CRISPR/Cas9-disrupted loci in human neural stem cells, specifically what fraction of disruption phenotypes became apparent after 4 cell divisions. Need distinguish enhancer loci, gene loci, and conserved region loci; report percentages and the paper title/PMID.'}

Step 3: TOOL

Files saved:

- /home/data4/xiangyu/BioDSA/tutorials/workdir/CRISPR_hNSC_phenotype_onset_summary.txt: Exact reported phenotype-onset percentages with quote, Table 1 verification, and study details.

- /home/data4/xiangyu/BioDSA/tutorials/workdir/CRISPR_neural_stem_cell_screen_phenotypes.csv: Initial search results identifying the paper.

Main findings:

The paper is **"Massively parallel disruption of enhancers active in human neural stem cells"** (Cell Reports 2024; PMID **38271204**; Geller et al., Noonan lab). sgRNA-Cas9 screens in human NSCs disrupted 10,674 genes and 26,385 conserved regions within 2,227 enhancers, measured at 4, 8, 12 cell divisions. Exact result: **61% of gene-disruption phenotypes** were already detected by 4 cell divisions, vs. only **30% of conserved-region (enhancer) disruption phenotypes**; enhancer phenotypes roughly doubled by 8 and again by 12 divisions. Note: the "enhancer loci" and "conserved region loci" are the same category here—the disrupted loci are conserved PhastCons elements inside enhancers; Table 1 confirms 1,385/2,263 = 61% (genes) and 355/1,175 = 30% (conserved enhancer regions).

Step 4: AI

Tool call: code_execution

Tool call input: {'language': 'python', 'code': "from pathlib import Path\np=Path('/home/data4/xiangyu/BioDSA/tutorials/workdir/CRISPR_hNSC_phenotype_onset_summary.txt')\nprint(p.read_text())"}

Step 5: TOOL

Executed Code:

```
from pathlib import Path
p=Path('/home/data4/xiangyu/BioDSA/tutorials/workdir/
CRISPR_hNSC_phenotype_onset_summary.txt')
print(p.read_text())
```

→ *Full code in Supplementary Materials (Code Execution #2)*

Output:

```
Paper: Massively parallel disruption of enhancers active in human neural stem
cells
Journal/Year: Cell Reports, 2024 (DOI: 10.1016/j.celrep.2024.113693)
PMID: 38271204
Authors/contact: Geller et al.; James P. Noonan lab (Yale)


... (24 more lines)
```

→ *See Code Execution #2 in Supplementary Materials for full implementation*

Step 6: AI

Tool call: add_to_graph

Tool call input: {'entities': [{'name': 'PMID:38271204', 'entity_type': 'PAPER', 'observations': ['Geller et al., Cell Reports 2024: massively parallel Cas9 disruption of enhancers active in human neural stem cells.', 'Pooled sgRNA screens measured phenotype onset at 4, 8, and 12 cell divisions.']}, {'name': 'hNSC Cas9 phenotype onset', 'entity_type': 'FINDING', 'observations': ['61% of gene-disruption phenotypes (1,385/2,263) were detected by 4 cell divisions.', '30% of conserved-region/enhancer-disruption phenotypes (355/1,175) were detected by 4 cell divisions; approximately 31% in rounded quiz wording.', 'Conserved regions are the disrupted loci within targeted enhancers.']}], 'relations': [{'from_entity': 'PMID:38271204', 'to_entity': 'hNSC Cas9 phenotype onset', 'relation_type': 'SUPPORTS'}]}

Step 7: TOOL

{"success": true, "results": {"entities_created": {"count": 2, "entities": [{"name": "PMID:38271204", "entityType": "PAPER", "observations": ["Geller et al., Cell Reports 2024: massively parallel Cas9 disruption of enhancers active in human neural stem cells.", "Pooled sgRNA screens measured phenotype onset at 4, 8, and 12 cell divisions."]}, {"name": "hNSC Cas9 phenotype onset", "entityType": "FINDING", "observations": ["61% of gene-disruption phenotypes (1,385/2,263) were detected by 4 cell divisions.", "30% of conserved-region/enhancer-disruption phenotypes (355/1,175) were detected by 4 cell divisions; approximately 31% in rounded quiz wording.", "Conserved regions are the disrupted loci within targeted enhancers."]}]}, "relations_created": {"count": 1, "relations": [{"from": "PMID:38271204", "to": "hNSC Cas9 phenotype onset", "relationType": "SUPPORTS"}]}}}

Step 8: AI

Answer

31% of enhancer loci showed disruption phenotypes after 4 cell divisions.

In the study (**PMID:38271204**), these enhancer targets were specifically **conserved regions within enhancers**: **355/1,175 phenotypes (≈30–31%)** were detected by 4 divisions. By contrast, **61% of gene-locus disruption phenotypes** were apparent by that time.

Thus, among the listed choices, select:

- **✔ 31% of enhancer loci**

- The equivalent mechanistic category is **31% conserved region loci**, since the disrupted enhancer loci were conserved PhastCons elements.

Results and Analysis

Summary

Answer

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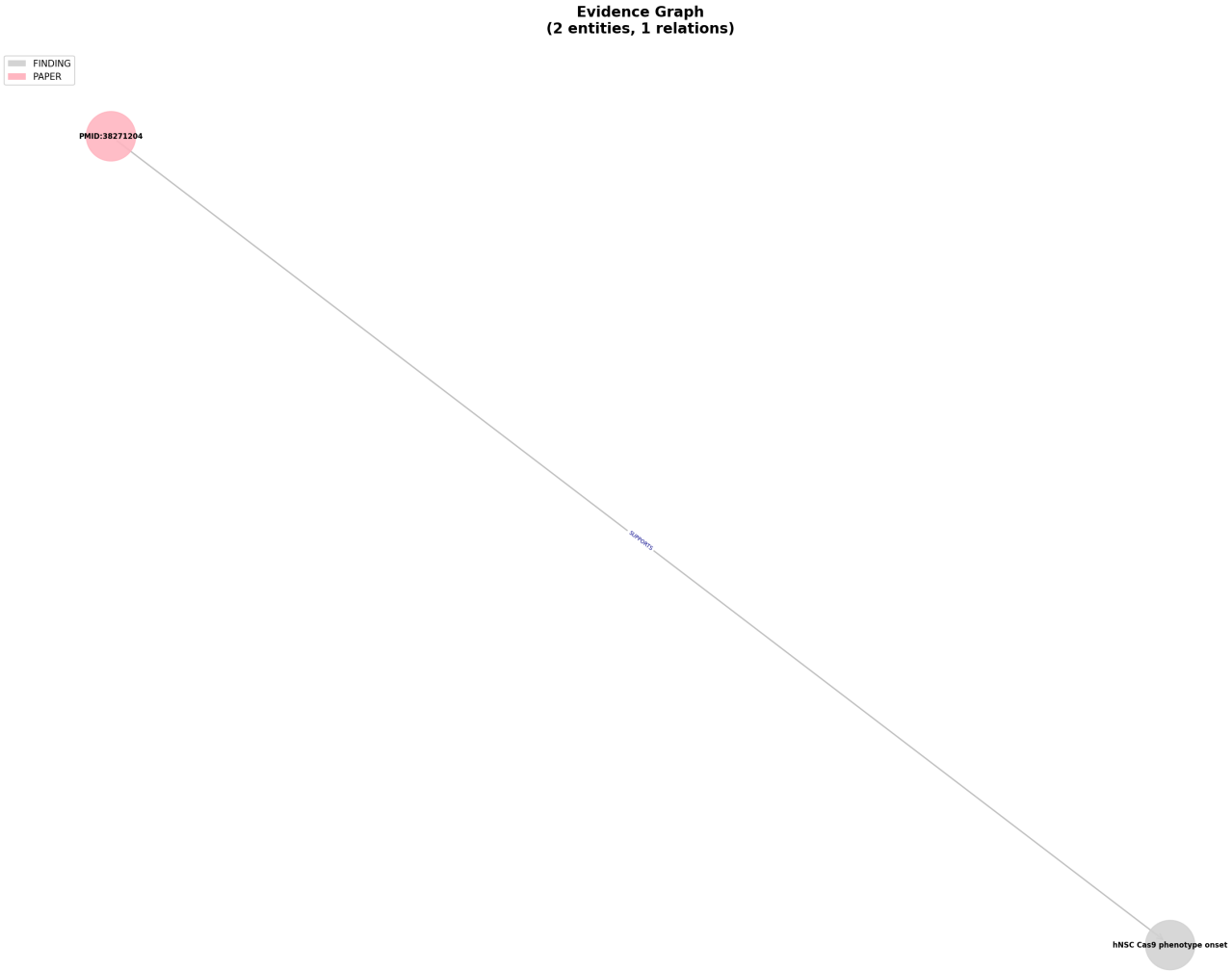
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Supplementary Materials

This section contains the evidence graph visualization and detailed code implementations.

Evidence Graph Visualization

The evidence graph contains **2 entities** and **1 relations** discovered during the research process. The graph shows the relationships between papers, chemicals, cell lines, genes, and research questions.



Supplementary Materials

This section contains detailed code implementations and execution results.